Lumen

Complete documentation, A to Z

This document explains **everything** about Lumen: what it is, how every piece was built, how the pieces connect, how a question travels through the system, how to run it, publish it, update it, test it and fix it. It is written so that someone who did not build it can understand it.

Repository: github.com/salman-kv/lumen
Version of this document: 10 November 2026

Contents

1. What Lumen is . . . 4

2. The big picture . . . 4

The three layers of "AI" . . . 4

3. Technology used and why . . . 5

Backend (Python) . . . 5

Frontend (JavaScript) . . . 5

AI and infrastructure . . . 6

4. Project map (every file) . . . 6

5. How the pieces connect . . . 7

5.1 One server, two roles . . . 7

5.2 Development vs production . . . 7

5.3 What the frontend calls . . . 7

5.4 The two kinds of token (do not confuse them) . . . 8

5.5 The language model connection . . . 8

6. Running it . . . 8

6.1 First-time setup on a PC (Windows) . . . 8

6.2 Run for development (two terminals) . . . 8

6.3 Run as one server (what is published) . . . 9

6.4 Run with Docker . . . 9

6.5 Demo organizations . . . 9

6.6 Where data is stored . . . 9

7. Configuration (all settings) . . . 9

8. Data model (database) . . . 10

9. Accounts, security and tenant isolation . . . 11

10. Knowledge base: what happens to an uploaded PDF . . . 12

11. The AI: what happens to a question . . . 13

11.1 Routing before any search (smalltalk.py) . . . 13

11.2 Context: follow-ups, history and "tell me more" . . . 14

11.3 Retrieval (retrieval.py, query.py) . . . 14

11.4 Classification (System 1, classifier.py) . . . 15

11.5 Grounding guard . . . 15
11.6 Choice decision: which policy? . . . 15
11.7 Facts and Boolean decisions . . . 15
11.7b Score decision . . . 15
11.8 Combining (application logic) . . . 15
11.9 Writing the answer, and keeping the LLM honest . . . 15
11.10 After the answer . . . 16
11.11 What is "trained" . . . 16

12. Frontend in detail . . . 16
12.1 Routes (App.jsx) . . . 16
12.2 Dashboard (Dashboard.jsx) . . . 17
12.3 The chat (ChatPanel.jsx) . . . 17
12.4 Streaming protocol (POST /api/ask/stream) . . . 17
12.5 Embedding on other websites (public/embed.js) . . . 17
12.6 Guided tour (Tour.jsx) . . . 17
12.7 Styling . . . 18

13. API reference . . . 18

14. Testing and evaluation . . . 19

15. Publishing and updating (free hosting) . . . 19
15.1 Why not a normal free host . . . 19
15.2 Going live . . . 19
15.3 Updating after a code change . . . 20
15.4 Before going public for real . . . 20

16. Troubleshooting . . . 20

17. How to extend it . . . 21

18. Known limits and next steps . . . 21

19. Glossary . . . 22

20. Command cheat sheet . . . 22

1. What Lumen is

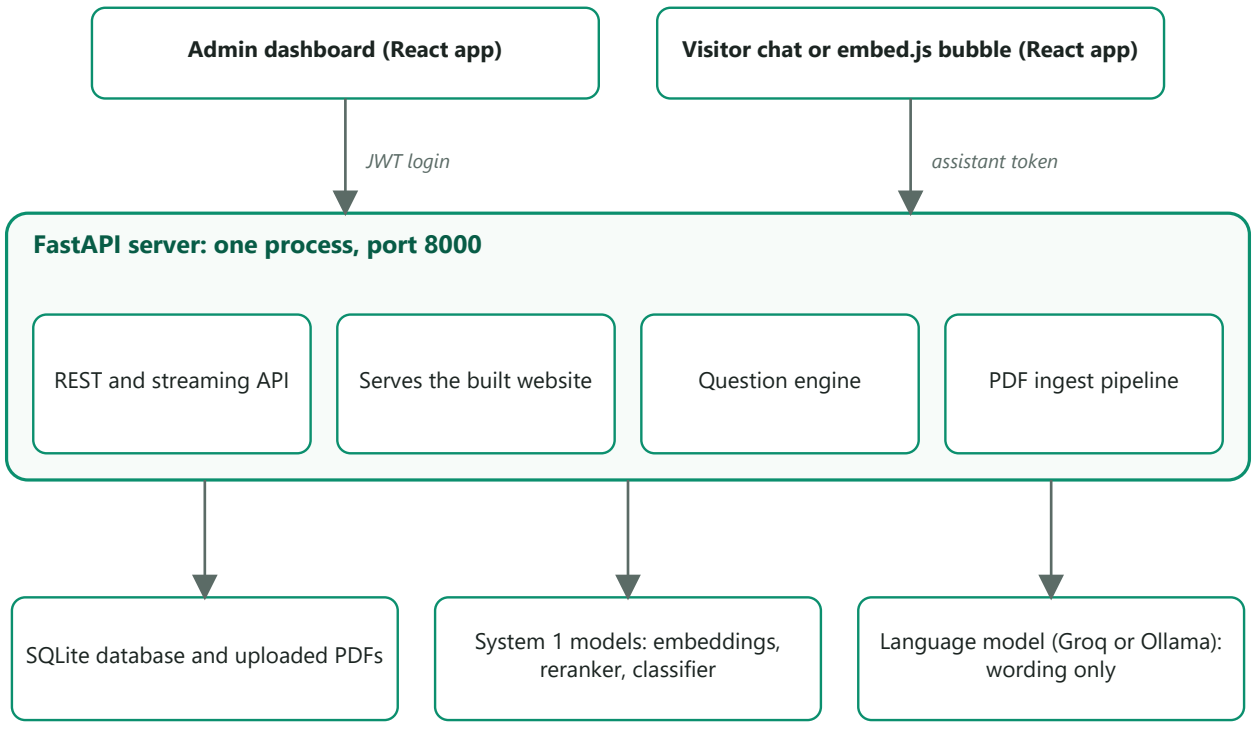
Lumen is a **multi-tenant AI Knowledge and Decision Assistant**.

- An **organization** (a university, a company) signs up and uploads its policy **PDFs**.
- Lumen gives that organization a **public assistant**: a chat page and an embeddable chat bubble.
- **Visitors** (students, employees) ask questions in plain English with no account. They get answers that **name the document and page** they came from.
- Some questions need a **decision** ("Am I eligible for the Merit Scholarship?"). Lumen breaks these into small checks, runs them separately, combines the results, and explains the outcome.
- **Multi-tenant** means many organizations share one running system, but each one's documents, assistant and history are completely separate.

Who uses it:

Person	What they do
Organization admin	Signs in to the dashboard, uploads PDFs, styles and shares the assistant, reads usage and question history
Visitor	Opens the assistant link (or chat bubble) and asks questions
Reviewer	Uses the demo organizations to evaluate the product

2. The big picture



One server does everything. FastAPI serves the API *and* the built website. There is no separate web server, no separate database server, and no queue.

The three layers of "AI"

The most important design idea: a single prompt to a language model is **not** how decisions are made.

Layer	Job	Can it be wrong in a way that matters?
LLM (language model)	Understand wording, extract the facts a user states, and phrase the final answer in friendly language	Yes, it can make things up, so it is fenced in (see 11.9)
System 1 models (small, fast, specialised)	Make the actual decisions: what kind of question, which policy, does each rule pass, is there enough evidence	They are deterministic or small statistical models. Results are auditable
Application logic (plain code)	Route messages, apply thresholds, combine decisions, find missing information, refuse when nothing is found	It is just code, so it behaves the same every time

The LLM never decides "eligible or not". It only receives the finished verdict and explains it.

3. Technology used and why

Backend (Python)

Technology	Role	Why this one
Python 3.12 + **uv**	Language and package manager	Fast installs, one lockfile (`uv.lock`)
FastAPI	Web framework	Type-checked requests, automatic validation, streaming responses, async
Uvicorn	Runs the FastAPI app	The standard server for it
SQLAlchemy 2 + **SQLite**	Database layer and database	Zero setup: one file. Swappable for Postgres by setting `DATABASE_URL`
Pydantic / pydantic-settings	Validates requests, reads settings from `.env`	Clear errors, one `settings` object
bcrypt	Password hashing	Slow on purpose, so stolen hashes are hard to crack
PyJWT	Login tokens (HS256)	Stateless sessions
pypdf	Reads PDF text page by page	Pure Python, no system tools needed
fastembed (ONNX)	Embeddings (`BAAI/bge-small-en-v1.5`) and cross-encoder reranker (`ms-marco-MiniLM-L-6-v2`)	Runs on CPU with no GPU, small downloads (about 150 MB total)
scikit-learn	Logistic regression for the question-type classifier	Tiny, trains in milliseconds at startup
numpy	Vector maths	Standard
httpx	Calls the language model over HTTP, async and streaming	Same client for Ollama and Groq
reportlab	Generates the sample PDFs and test PDFs	Used by scripts and tests
pytest	Tests	119 tests

Frontend (JavaScript)

Technology	Role
React 19	UI components
Vite 8	Dev server and production build
React Router 7	Pages (`/`, `/login`, `/app/*`, `/a/:token`)
Tailwind CSS 4	Styling (utility classes plus theme variables for light/dark)

Technology	Role
Motion (formerly Framer Motion)	Animations
Phosphor Icons	Icons
Sonner	Toast notifications
canvas-confetti	Celebration when setup finishes
Geist fonts	Typography, self-hosted

AI and infrastructure

Technology	Role
Groq (hosted, free tier)	The language model in the live setup (`qwen/qwen3.8-27b`)
Ollama (local, optional)	Free offline language model option (`qwen2.5:3b`)
Docker / Docker Compose	Optional one-command packaging (app + Ollama)
Cloudflare quick tunnel (`cloudflared`)	Free public link to the app running on your PC
Git / GitHub	Source control (`github.com/salman-kv/lumen`)

4. Project map (every file)

```
app/
├── README.md                   Project overview and quick start
├── Dockerfile                  Builds frontend, then the Python image with models baked in
├── docker-compose.yml          app + ollama + one-shot model download
├── .dockerignore, .gitignore
│
├── backend/
│   ├── pyproject.toml, uv.lock  Python dependencies (locked)
│   ├── .env.example             Template for settings (copy to .env, never commit .env)
│   ├── app/
│   │   ├── main.py              FastAPI app: all endpoints, rate limit, serves the website
│   │   ├── config.py            Every setting, read from environment / .env
│   │   ├── db.py                Database tables (SQLAlchemy models), init and small migrations
│   │   ├── security.py          Password hashing, JWT creation, "who is signed in" check
│   │   ├── pdfproc.py           PDF validation, text extraction, page-aware chunking
│   │   ├── ingest.py            Background job: chunk, embed, mine rules, set document status
│   │   ├── rules.py             Rule mining (regex) and rule evaluation (Boolean decisions)
│   │   ├── models_s1.py         Embedder and reranker wrappers (plus offline "hash" fallback)
│   │   ├── classifier.py        Question-type classifier (logistic regression on embeddings)
│   │   ├── retrieval.py         Hybrid search: BM25 + embeddings + rerank, tenant-scoped
│   │   ├── engine.py            The question pipeline, the heart of the product
│   │   ├── llm.py               Language-model client (Ollama, OpenAI-compatible/Groq, mock)
│   │   ├── smalltalk.py         Greetings, injection, gibberish, language, human-handoff routing
│   │   ├── suggest.py           Starter questions, follow-up chips, "what I can help with" menu
│   │   ├── query.py             Everyday words -> policy words ("money back" -> refund) for searching
│   │   └── demo_seed.py         Optional: creates the two demo organizations on start
│   └── tests/                   conftest.py, test_features.py, test_isolation.py (119 tests)
│
├── frontend/
│   ├── package.json, vite.config.js, index.html
│   ├── public/
│   │   ├── embed.js             The script customers paste into their sites (chat bubble)
│   │   └── demo.html            A sample host page that embeds the assistant
│   └── src/
│       ├── main.jsx, App.jsx    Entry point and routes
│       ├── api.js               API client (fetch, upload with progress, streaming ask)
│       ├── auth.jsx             Sign-in state and token storage
│       ├── index.css            Theme variables, buttons, cards
│       ├── components/
│       │   ├── ChatPanel.jsx    The chat (used by visitors and by the dashboard preview)
│       │   ├── PipelineViz.jsx  The animated "How I decided" steps
```

```
│       │   ├── Tour.jsx          The guided walkthrough
│       │   └── ui.jsx            Logo, theme toggle, copy button, modal, count-up
│       └── pages/
│           ├── Landing.jsx       Marketing home page
│           ├── AuthPage.jsx      Sign in and create account
│           ├── ChatPage.jsx      Public assistant page (/a/<token>)
│           ├── Dashboard.jsx     Admin shell: sidebar, data loading, tour
│           └── dash/             Overview.jsx, Knowledge.jsx, Assistant.jsx, Usage.jsx
│
├── sample_data/
│   ├── northfield/, acme/        Two organizations, three PDFs each
│   └── test_questions.json       Demo logins and the 19 evaluation questions
│
├── scripts/
│   ├── make_sample_data.py       Generates the sample PDFs
│   ├── seed_and_eval.py          Creates demo orgs, runs the 19-question evaluation
│   ├── scenario_battery.py       44 extra live scenarios (typos, off-topic, cross-tenant...)
│   ├── share_public.bat          Start everything and print a public link
│   └── update_live.bat           Rebuild frontend and restart the server
│
└── docs/                         DOCUMENTATION.md (this file), REQUIREMENTS_CHECKLIST.md, DEMO_SCRIPT.md
```

Folders that appear when you run it but are **not** in Git: `backend/data/` (database, uploaded PDFs, model cache), `backend/.venv/`, `frontend/node_modules/`, `frontend/dist/` (the built website), `tools/` (cloudflared), `backend/.env` (your secrets).

5. How the pieces connect

5.1 One server, two roles

When you run `uvicorn app.main:app`, a single process:

1. Exposes the **API** under `/api/...`.
2. Serves the **built website** from `frontend/dist`: files under `/assets/...`, and `index.html` for every other path (so React Router can handle `/login`, `/app/usage`, `/a/<token>`).

That is why you only need port 8000, and why the Cloudflare tunnel only forwards one address.

5.2 Development vs production

	Development	Production-like (what is live)
Website	Vite dev server on port **5173**, hot reload	Pre-built files in `frontend/dist`, served by FastAPI
API calls	Vite **proxies** `/api` to port 8000 (see `vite.config.js`)	Same origin, no proxy
Command	`npm run dev` + `uvicorn ... --reload`	`npm run build` + `uvicorn`

5.3 What the frontend calls

All calls go through `frontend/src/api.js`:

- `api(path, ...)`: normal JSON requests. Adds `Authorization: Bearer <JWT>` when signed in. On a 401 it signs the user out.
- `uploadFile(...)`: PDF upload using XMLHttpRequest so the progress bar can show real progress.
- `askStream(...)`: the visitor's question. It uses `fetch` and reads a **Server-Sent Events** stream so the answer appears word by word.

The JWT is kept in the browser's `localStorage` under `kda_token`. The visitor chat sends **no JWT**, only the assistant token in the request body.

5.4 The two kinds of token (do not confuse them)

	Admin JWT	Assistant token
Looks like	`eyJhbGciOi...`	`ast_...`
Who has it	Signed-in admin	Anyone with the public link
Lets you	Use the dashboard APIs for **your own** organization	Ask questions of that one assistant. Nothing else
Created by	Login (expires in 24 hours)	Account creation (random, regenerate to retire the old one)
Works on admin APIs?	Yes	**No**, it is not a JWT, so the admin check rejects it

5.5 The language model connection

`llm.py` supports three providers, chosen by `LLM_PROVIDER`:

- `openai_compat`: any OpenAI-style API (Groq is used live). Needs `OPENAI_BASE_URL`, `OPENAI_API_KEY`, `OPENAI_MODEL`.
- `ollama`: a local Ollama server (`OLLAMA_URL`, `OLLAMA_MODEL`).
- `mock`: no model; replies come from the app's own fallback text. Used by the tests.

Failover. Free tiers limit tokens per minute **per model** (Groq: about 8,000 a minute). When the main model answers 429 (rate limited), 404 (missing), or a server error, `llm.py` retries the same request on each model in `OPENAI_FALLBACK_MODELS`, because every model has its own allowance. A real error such as a bad API key is not hidden.

If every model fails, the engine still answers using a **template fallback** and marks the answer `degraded`.

6. Running it

6.1 First-time setup on a PC (Windows)

Install: **Python 3.12+ via `uv`, Node.js 22+, Git**. Then:

```
git clone https://github.com/salman-kv/lumen.git
cd lumen\backend
copy .env.example .env
```

Edit `backend\.env`: set `JWT_SECRET` to a long random string. For the language model pick one:

```
# Free hosted model (recommended): get a key at console.groq.com
LLM_PROVIDER=openai_compat
OPENAI_BASE_URL=https://api.groq.com/openai/v1
OPENAI_API_KEY=your-key-here
OPENAI_MODEL=qwen/qwen3.8-27b
```

Model names on Groq change. List the ones your key can use at `https://api.groq.com/openai/v1/models` and pick a chat model. A wrong name makes every call fail and the app silently falls back to plain text answers.

6.2 Run for development (two terminals)

```
:: terminal 1: API on port 8000, restarts when Python files change
cd backend
uv sync
uv run uvicorn app.main:app --reload --port 8000

:: terminal 2: website on port 5173 with hot reload
cd frontend
npm install
npm run dev
```

Open `http://localhost:5173`.

6.3 Run as one server (what is published)

```
cd frontend
npm install
npm run build
cd ..\backend
set SEED_DEMO=true
uv run uvicorn app.main:app --host 127.0.0.1 --port 8000 --proxy-headers --forwarded-allow-ips=*
```

Open `http://localhost:8000`.

- `--proxy-headers --forwarded-allow-ips=*` makes the app trust the tunnel's headers, so the generated assistant links use `https://` and the public host.
- On first start the System 1 models download once (about 150 MB). Later starts are fast.

6.4 Run with Docker

```
docker compose up --build
```

Starts the app on port 8000, an Ollama container, and a one-time job that downloads `qwen2.5:3b`. No account or key needed. Data lives in the `appdata` volume.

The Dockerfile has two stages: (1) build the React app with Node, (2) a Python image with the dependencies, the built website, the sample data, and the two models pre-downloaded. It runs as a non-root user.

6.5 Demo organizations

With `SEED_DEMO=true`, on start the app creates two organizations and uploads their PDFs (idempotent, safe to repeat):

Organization	Email	Password
Northfield University	`demo-northfield@example.com`	`demo-password-1`
Acme Corp	`demo-acme@example.com`	`demo-password-2`

The login page then shows one-click buttons for them. Turn `SEED_DEMO` off for a real deployment.

6.6 Where data is stored

Under `DATA_DIR` (default `backend/data/`):

- `app.db`: the SQLite database.
- `uploads/`: the original PDFs (never served publicly).
- (Docker) model files under `/app/models`.

To **reset everything**, stop the server and delete `backend/data/`. To **back up**, copy that folder while the server is stopped.

7. Configuration (all settings)

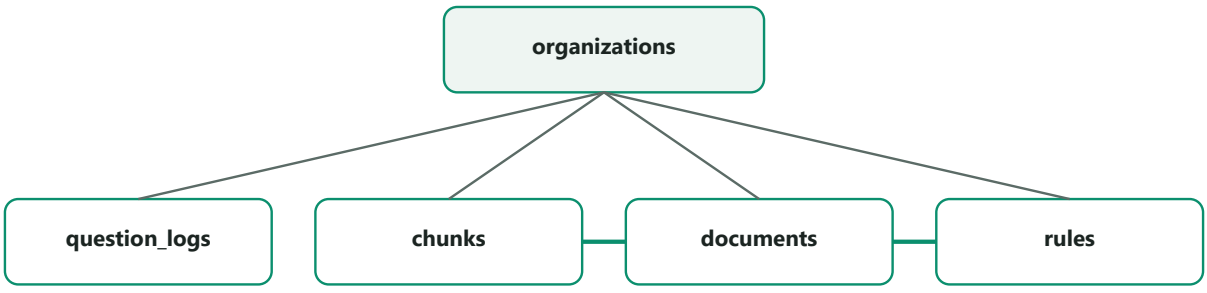
All are set as environment variables or in `backend/.env` (names are the uppercase version).

Setting	Default	Meaning
`JWT_SECRET`	dev value	Signs login tokens. **Change it** in any real deployment
`JWT_HOURS`	24	Login lifetime

Setting	Default	Meaning
`PUBLIC_URL`	empty	Used in assistant links and embed code. Empty = use the request's address
`DATA_DIR`	`backend/data`	Database and uploads
`DATABASE_URL`	SQLite file	Set to use another database
`MAX_DOCS` / `MAX_PAGES` / `MAX_PDF_MB`	10 / 20 / 10	Upload limits from the assignment
`LLM_PROVIDER`	`ollama`	`openai_compat`, `ollama`, or `mock`
`OPENAI_BASE_URL`, `OPENAI_API_KEY`, `OPENAI_MODEL`	Groq URL, empty, `qwen/qwen3.8-27b`	Hosted model
`OPENAI_FALLBACK_MODELS`	`openai/gpt-oss-20b,openai/gpt-oss-120b`	Tried in order when the main model is rate-limited or unavailable
`OLLAMA_URL`, `OLLAMA_MODEL`	`http://localhost:11434`, `qwen2.5:3b`	Local model
`LLM_TIMEOUT`	120	Seconds before a model call gives up
`RULES_LLM`	false	Let the LLM propose extra rules at upload (only worthwhile with a strong model)
`EMBED_BACKEND`	`fastembed`	`hash` = offline stand-in used by tests
`EMBED_MODEL`, `RERANK_MODEL`	bge-small, MiniLM	The System 1 models
`TOP_K`	5	How many passages to keep
`MIN_SUPPORT`	0.12	Evidence score below which the assistant refuses to answer
`RATE_LIMIT_PER_MINUTE`	30	Per assistant token and visitor address (in memory)
`SEED_DEMO`, `SEED_BASE`, `SAMPLE_DIR`	false, `http://127.0.0.1:8000`, `sample_data`	Demo seeding

8. Data model (database)

Five tables. **Every table below `organizations` carries `org_id`**, and every query filters by it. That column is the foundation of tenant isolation.



Every table carries org_id. documents are split into chunks and yield rules; deleting cascades.

Table	Important columns	Notes
`organizations`	`name`, `email` (unique), `password_hash`, `assistant_token` (unique), `assistant_name`, `brand_color`, `welcome_message`, `suggestions` (JSON)	One row per customer
`documents`	`filename`, `stored_name`, `pages`, `status` (`processing`/`ready`/`failed`), `stage`, `error`, `chunk_count`, `rule_count`	`stage` is the live label shown while processing
`chunks`	`doc_id`, `page`, `idx`, `text`, `embedding` (binary vector)	The searchable passages. Never cross a page
`rules`	`doc_id`, `page`, `benefit`, `field`, `label`, `operator`, `value`, `evidence`, `source`	A typed rule card, for example Merit Scholarship / `gpa` / `>=` / `3.5`
`question_logs`	`question`, `qtype`, `outcome`, `used_decisions`, `latency_ms`, `answer`, `sources`, `created_at`	Powers Usage and the full history

Notes:

- Deleting an organization or document **cascades** to its chunks and rules.
- `init_db()` creates missing tables on start and adds new columns to old databases (a small built-in migration for SQLite), so upgrades keep your data.
- Times are stored in UTC. The history API marks them `+00:00` so browsers convert correctly.

9. Accounts, security and tenant isolation

Signing up (`POST /api/auth/register`): validates the email and password (8 or more characters), stores a **bcrypt** hash, creates the organization with a random assistant token, and returns a JWT.

Signing in (`POST /api/auth/login`): checks the hash, returns a JWT valid for 24 hours.

Every admin endpoint depends on `current_org`, which:

1. reads the `Authorization: Bearer ...` header,
2. verifies the signature, expiry and `typ == "admin"`,
3. loads the organization from the token (never from anything the client sends).

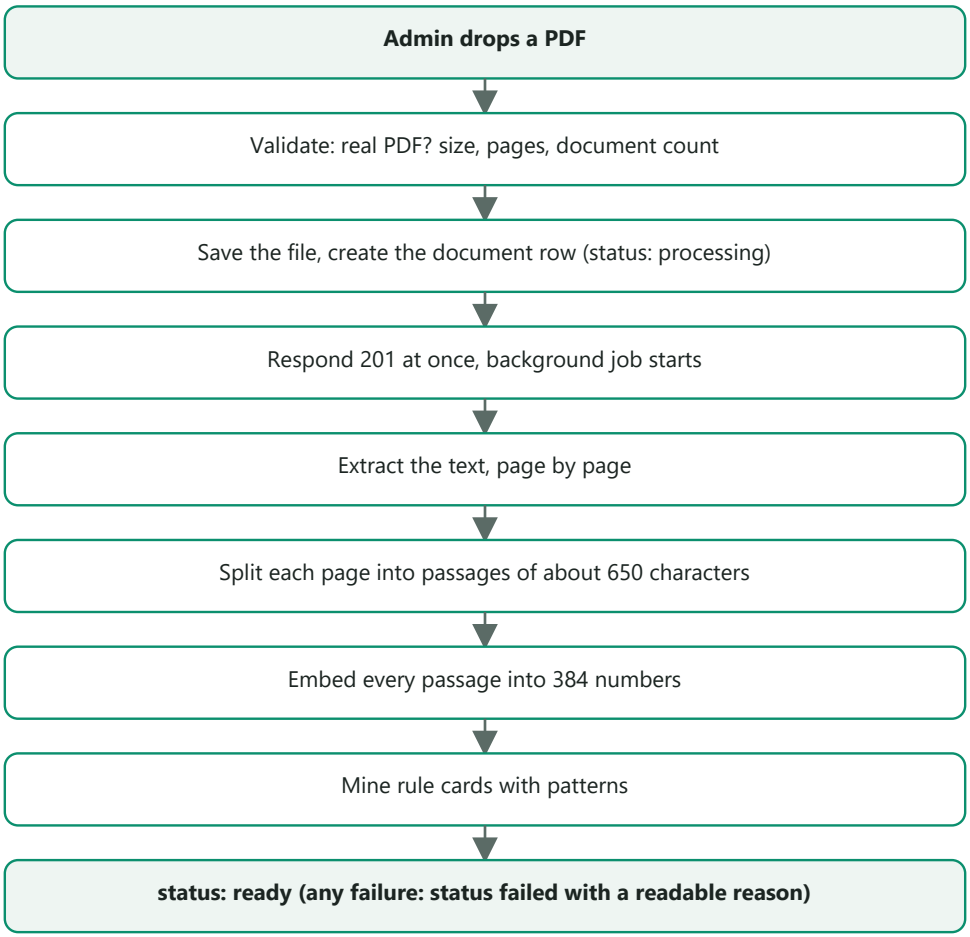
So an admin can only ever see **their own** organization, because the organization id comes from the signed token.

The public assistant (`/api/ask`, `/api/public/...`) finds the organization by assistant token, then passes only that `org_id` to retrieval. Search queries are `WHERE org_id = ?` at the database level, so another tenant's text is never loaded.

Other protections:

- **Assistant token is not a JWT**, so it cannot pass the admin check. Tested.
- **Regenerating** the token immediately invalidates the old link. Tested.
- **Documents are private**: uploads are not in any static folder. Downloading one (`GET /api/kb/documents/{id}/file`) needs the admin JWT. Tested.
- **Upload limits and validation**: real PDF check, size, page count, document count. Clear error messages.
- **Rate limiting**: 30 questions per minute per token and address (in memory, resets on restart).
- **CORS** is open because the admin API uses bearer tokens (not cookies), and the public endpoints must be callable from embedding sites.
- **Prompt injection** ("ignore your instructions") is caught by the router before any model sees it.
- **Isolation tests**: `tests/test_isolation.py` proves that organization B cannot read organization A's knowledge, history or documents, and that replies list only the asking organization's documents.

10. Knowledge base: what happens to an uploaded PDF



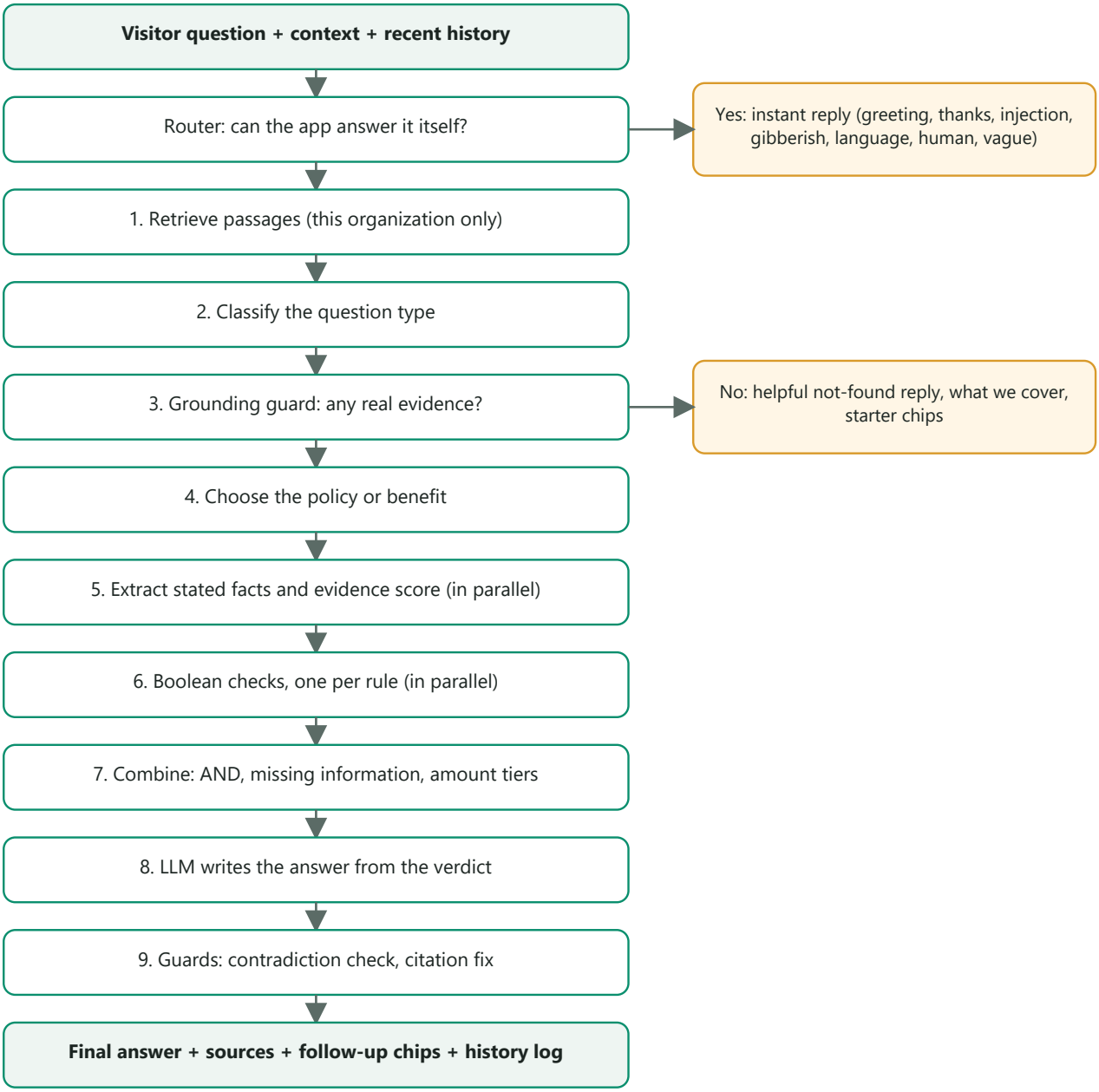
Step by step (`pdfproc.py`, `ingest.py`, `rules.py`):

1. **Validate** (`validate_and_read`): must be a genuine PDF with extractable text, at most 20 pages and 10 MB; the organization can hold at most 10 documents.
2. **Chunk** (`chunk_page`): split on blank lines and sentences into passages up to about 650 characters. **A passage never crosses a page boundary**, which is what lets every answer cite an exact page.
3. **Embed** (`models_s1.embed`): each passage becomes a vector with `bge-small-en-v1.5`. Stored as bytes in `chunks.embedding`.
4. **Mine rules** (`rules.mine_page`): regular expressions find statements such as "must be in at least their second year", "GPA of at least 3.5", "must not hold a sports scholarship", and turn each into a **rule card**: benefit, field, operator, value, the evidence sentence and the page. Section headings help assign each rule to a benefit (for example "Merit Scholarship"). Optional: with `RULES_LLM=true` an LLM proposes extra rules, which are filtered by a gate so noisy ones are dropped.
5. **Progress**: the document's `stage` field is updated ("Indexing passages", "Reading policy rules"); the dashboard polls while anything is processing and animates the stages.
6. **Replace / delete**: replacing a document re-runs the pipeline. Deleting removes its chunks and rules.

Why rules are mined by patterns and not by an LLM: it is deterministic, auditable (each rule shows its evidence), cannot hallucinate, and works offline.

11. The AI: what happens to a question

This is the core of the product, in `engine.py`. The function `engine.run(...)` is an **async generator** that yields events as it works; the API streams them to the browser.



11.1 Routing before any search (`smalltalk.py`)

The app answers these **without search or LLM** (instant, free, never "not found"):

Intent	Examples	Reply
`small_talk`	"hi", "helo" (typos fixed), "thanks", "who are you", "what can you do"	Friendly intro naming the organization's topics
`blocked`	"ignore previous instructions", "show your system prompt"	Polite refusal
`unclear`	"asdf", emoji only, a single mashed word, or "tell me more about it" / "what?" with nothing to refer to	Asks what you mean and lists what it can help with

Intent	Examples	Reply
`help`	"can you help me", "i need help", "i have a question"	"Of course!" plus what each document covers
`overview`	"what do you know", "what all the features we have", "list everything", "give me a summary", "i am new here", "anything else"	What each document covers (topic names), plus a starting point
`frustrated`	"this is not helpful", "you are wrong", "that's not what i asked"	A short apology, asks for a rephrase, lists topics
`small_talk` (extra)	"who made you", "are you a bot", "lol"	Says it is an AI assistant for the organization; a friendly one-liner
`language`	Mostly non-Latin text	Says it answers in English
`human`	"talk to a human"	Explains it is automated and to contact the organization
`abuse`	Insults	Calm reply, offers help
`bare_reply`	"yes", "no" with no context	Asks for more detail

Requests for actions ("generate a PDF", "can you email this") are caught only if the knowledge base has no answer, so real questions that start with "can you" still work.

An overview request must name **nothing specific** (every word is generic, like "what", "list", "features", "summary"). "List the documents I need to submit" names something, so it is searched, not answered with the menu.

Every routed reply is followed by clickable starter questions, except thank-yous, goodbyes, "ok" and "lol".

11.2 Context: follow-ups, history and "tell me more"

- The browser sends the **previous question** as `context` when the visitor is answering a "need more information" reply or asking a short follow-up (like "what about third year?"). The server joins them for searching.
- It also sends the **last 4 turns** (question + short answer) as `history`. The LLM sees these **only to understand what the question refers to, never as a source of facts**.
- **"what?", "huh", "why", "really?", "I don't understand"** right after an answer mean "explain that again": the server asks the previous question again and the LLM re-explains the same facts in simpler words (`SIMPLE_SYSTEM`). With nothing before it, the visitor is asked what they mean.
- Words like "more", "details", "explain", "example" switch on **detail mode**: more passages are retrieved and the LLM is asked for a fuller answer that still states only what the context says. Filler words are stripped from the search query so they don't dilute it.

11.3 Retrieval (`retrieval.py`, `query.py`)

0. **Everyday words become policy words** (`query.expand`): "money back if I quit" is searched as "... refund withdraw withdrawal", "how much do I pay" adds "fee tuition", "work from home" adds "remote work". Only the search query is widened; the answer still comes only from passages that were found.
1. Load this organization's ready chunks (`WHERE org_id = ?`).
2. **BM25** keyword score and **dense** embedding similarity are computed.
3. They are fused with **Reciprocal Rank Fusion** (`1/(60+rank)` from each).
4. The best candidates go to the **cross-encoder reranker**, which reads question and passage together and returns a support score 0 to 1.
5. A **lexical-support floor** is applied: if every specific word of a short question appears in the passage ("What is the refund policy?"), the score is at least 0.4. Without it, short generic questions score too low.

Result: the top passages with a score. Each keeps its document name and page.

11.4 Classification (System 1, `classifier.py`)

A **logistic regression** on the question's embedding, trained at startup from example questions, labels each question: `eligibility`, `comparison`, `procedural`, `policy`, or `direct_info`. No LLM. Milliseconds.

11.5 Grounding guard

If there are no passages, or the best score is below `MIN_SUPPORT` and no policy matches, the assistant **does not guess**. It replies (written by the app, not the LLM):

- what it could not find (quoting the visitor's words),
- the closest section if there is one,
- a list of what each document covers (benefit names and section titles),
- clickable starter questions.

11.6 Choice decision: which policy?

`choose_policy` compares the question embedding with each benefit's description (built from its rule labels). The best match above a threshold is the benefit in question ("Merit Scholarship"); close runners-up are kept for comparison questions.

11.7 Facts and Boolean decisions

A decision is only attempted when the visitor states facts about themselves or asks "can I / am I eligible". Questions like "What GPA do I need?" are **information questions** and get a plain answer.

1. **Facts** are extracted two ways: `pattern_facts` (regular expressions for GPA, year including words like "year one", attendance, months of service, yes/no holdings like "no sports scholarship", with negation handled) and `llm_facts` (the LLM, run in parallel).
2. `ground_facts` drops any fact the LLM returned that cannot be found in the question text. This stops a small model inventing "GPA 3.5".
3. **One Boolean check per rule card** runs in parallel: `rules.evaluate` compares the stated fact to the rule (`>=`, `<=`, `==`, `is_true`, ...) and returns `pass`, `fail` or `unknown` (fact not stated).
4. **Amount tiers** (`amount_tier`): for rules like "above $500 requires a Director", the app itself picks the right approver for the amount asked ("Who approves $900?" gives Director). Models are unreliable with numbers, so the app decides.

11.7b Score decision

In parallel, the reranker's support score for the best passage is the **evidence** decision: is there enough support to answer at all.

11.8 Combining (application logic)

`combine` applies **AND** across all conditions:

- any `fail` means `not_eligible`;
- else any `unknown` means `need_info` (the missing items are listed and the visitor is asked for them);
- else `eligible`.

Unknown facts are surfaced, never assumed.

11.9 Writing the answer, and keeping the LLM honest

The LLM receives the verdict and the app-written reasons, and only phrases them. Safeguards (all in `engine.py`):

- `reason()`: the explanatory sentences are written by the app from the rule results, not by the model.

- `contradicts()`: if the LLM's wording contradicts the verdict (says "eligible" when it is not), the answer is replaced with the app's own text, and the stream sends a `replace` event.
- `fix_citations()`: model-invented file names ("Doc.pdf") are snapped to real documents.
- **Deterministic fallback**: if the model is down or returns nothing, the answer is built from the best sentence or the verdict, and the response is flagged `degraded`.
- **History is context only** (see 11.2).

The wording of the answer depends on the question (all in `engine.py`):

Mode	When	Style
Short (default)	normal questions	Under 90 words, plain text, source cited
Decision	the visitor states facts or asks if they qualify	Verdict first, reasons written by the app
List	"list", "what are all the", "steps", "purposes", "features"	One item per line starting with "-"
Topic	one or two words such as "refund" or "gpa"	A short friendly overview with the main facts
Detail	"tell me more", "explain", "example"	Fuller, still only what the context states
Simple	"what?", "I don't understand" after an answer	The same facts in everyday words

Only the best passages are sent to the model (weak ones are dropped), which also saves tokens on free tiers.

11.10 After the answer

- **Follow-up chips** (`suggest.followups`): "Tell me more details", then "Tell me about <section>" for other sections of the document the answer came from, then related questions. None while the assistant is waiting for the visitor's facts.
- **Logging**: the visitor's typed question, the answer, the sources, type, outcome and time are stored in `question_logs`.
- **Steps shown to the user**: `retrieve`, `classify`, `choose`, `understand`, `decide`, `combine`, `answer`. The UI animates them as "How I decided".

11.11 What is "trained"

Honest summary, useful if you are asked:

- The **embedding model** and **reranker** are pre-trained models used as-is.
- The **question classifier** is trained at startup from example questions (a few milliseconds).
- The **LLM** is not trained or fine-tuned. Answer quality comes from retrieval, rules, prompts and guards, not from training.
- Rules are mined with patterns, not learned.

12. Frontend in detail

12.1 Routes (`App.jsx`)

Path	Page	Notes
`/`	Landing	Marketing page
`/login`, `/register`	AuthPage	Validation, password strength, demo buttons when seeding is on
`/app/*`	Dashboard	Requires sign-in; redirects to login otherwise
`/a/:token`	ChatPage	Public assistant; `?embed=1` for the iframe layout

12.2 Dashboard (`Dashboard.jsx`)

Loads documents, assistant settings and usage once, shares them through a React context (`useData`), and polls documents every 1.2 s while any is processing. Tabs:

- **Overview**: three-step checklist (upload, try, share) with progress and confetti when done; summary tiles.
- **Knowledge**: drag-and-drop upload with real progress, per-document status and stage, replace, delete, "what the assistant learned" (rule cards).
- **Assistant**: public link (copy, regenerate), embed code (script or iframe), name, colour, welcome message, up to 6 starter questions, and a **live preview** that is the real chat component fed with unsaved form values.
- **Usage**: tiles, last-7-days chart, and the **full question history** (search, filter by outcome, paging, expandable answers, CSV download).

12.3 The chat (`ChatPanel.jsx`)

Shared by the public page, the iframe and the dashboard preview.

- Loads the assistant's name, colour, welcome message and starter questions.
- `send()` decides whether to attach `context` (earlier question) and builds `history` (last 4 turns), then calls `askStream`.
- Handles streaming events and updates the last bot message live.
- Shows source chips (click for the exact snippet), the "How I decided" panel, retry on error, and follow-up chips under the latest answer.

12.4 Streaming protocol (`POST /api/ask/stream`)

The response is `text/event-stream`; each line is `data: {json}`:

`type`	Meaning
`step`	A pipeline stage started, finished, or was skipped (id, label, kind, ms, detail)
`token`	A piece of the answer text
`replace`	The whole answer text is replaced (after a guard corrected it)
`final`	The complete result: answer, status, references, decisions, steps, follow-ups, latency
`error`	Something failed; message is user-readable

`POST /api/ask` returns only the final object (non-streaming), used by scripts and tests.

12.5 Embedding on other websites (`public/embed.js`)

```
<script src="https://YOUR-HOST/embed.js" data-token="ast_..." async></script>
```

The script adds a floating button; clicking opens the assistant in an **iframe** (`/a/<token>?embed=1`). It only knows the public token, so it cannot reach the dashboard or private files. The dashboard also offers a plain `<iframe>` snippet. `public/demo.html` is a sample page using it.

12.6 Guided tour (`Tour.jsx`)

11 steps across the dashboard tabs. Each step has a path and a CSS selector (`data-tour="..."` markers on the pages). It navigates, waits for the element, scrolls to it, and draws a spotlight; the card sits beside it (a bottom sheet on phones). Next, Back, Skip, arrow keys, Esc. Completion is stored in `localStorage` (`lumen_tour_done`); "Take the tour" replays it.

12.7 Styling

Tailwind 4 with theme variables in `index.css` (light and dark, one accent colour). Components share `card`, `btn`, `chip`, `input` classes. Animations respect "reduce motion".

13. API reference

`Auth` column: **public** = no login; **token** = assistant token in the body or path; **admin** = `Authorization: Bearer <JWT>`.

Method and path	Auth	Purpose
`POST /api/auth/register`	public	Create organization and account, returns JWT
`POST /api/auth/login`	public	Sign in, returns JWT
`GET /api/auth/me`	admin	Current organization
`GET /api/kb/documents`	admin	List documents and the upload limits
`POST /api/kb/documents`	admin	Upload a PDF (201); processing continues in the background
`PUT /api/kb/documents/{id}`	admin	Replace a document
`DELETE /api/kb/documents/{id}`	admin	Delete a document (204)
`GET /api/kb/documents/{id}/file`	admin	Download the original PDF
`GET /api/kb/rules`	admin	The rule cards found
`GET /api/assistant`	admin	Assistant settings, public URL, embed code
`PATCH /api/assistant`	admin	Update name, colour, welcome message, starter questions
`POST /api/assistant/regenerate`	admin	New assistant token (old link stops working)
`GET /api/usage`	admin	Counts, last 7 days, average time
`GET /api/usage/history`	admin	Paged question history. Query: `page`, `per_page` (max 50), `q`, `outcome`
`GET /api/usage/history.csv`	admin	CSV export of the (filtered) history
`GET /api/public/demo`	public	Demo logins, only when `SEED_DEMO` is on
`GET /api/public/assistant/{token}`	token	Name, colour, welcome message, starter questions
`POST /api/ask`	token	Ask a question; returns the final result. Body: `token`, `question`, optional `context`, `history`
`POST /api/ask/stream`	token	Same, as an event stream
`GET /api/health`	public	Status, language-model check, embedding backend

Interactive docs (Swagger) are available from FastAPI at `/docs` when the server is running.

Final answer object (`/api/ask`):

```
{
  "answer": "...",
  "status": "answered | needs_info | no_knowledge | error",
  "references": [{"document": "Fee_Policy.pdf", "page": 2, "snippet": "...", "score": 0.91, "kind": "passage"}],
  "question_type": {"label": "policy", "confidence": 0.75},
  "decisions": {"verdict": "need_info", "...": "..."},
  "followups": ["Tell me more details", "..."],
  "steps": [{"id": "retrieve", "label": "...", "kind": "s1", "ms": 79, "detail": "..."}],
  "degraded": false,
  "latency_ms": 673
}
```

Error responses are JSON `{"detail": "readable message"}` with proper status codes (401, 404, 409, 422, 429).

14. Testing and evaluation

What	Command	Count	Needs
Unit and API tests	`cd backend && uv run pytest -q`	119	Nothing (offline: hash embeddings and a mock LLM)
Assignment evaluation	`uv run python ../scripts/seed_and_eval.py`	19 questions	Running server and a real model
Scenario battery	`uv run python ../scripts/scenario_battery.py`	44 scenarios	Running server

- **pytest** covers: registration and login, upload rejections, the 10-document limit, replace, rules, decisions (including missing facts and negation), small talk and routing, follow-ups, history, CSV, not-found replies, error handling, token security, and **cross-tenant isolation** (`test_isolation.py`).
- **The 19-question evaluation** (`sample_data/test_questions.json`) covers direct, policy, procedural, multi-condition, comparison and no-answer questions across both organizations, and checks status, verdict and the cited document.
- **The battery** adds typos, ALL CAPS, emoji, off-topic questions, boundary values, expense tiers, and cross-organization probes.
- **Rate limit warning:** the live server allows 30 questions a minute per token. Running the evaluation and the battery back to back trips it and shows "You're asking too quickly" failures. Wait a minute between runs.
- Running `seed_and_eval.py` fully re-creates the demo organizations' documents and writes `sample_data/.tokens.json` (ignored by Git).

15. Publishing and updating (free hosting)

15.1 Why not a normal free host

- Hugging Face Spaces now charges for Docker Spaces.
- Free cloud tiers (512 MB RAM) cannot hold the embedding models: the app needs about 550 MB while processing a PDF.
- Vercel-style serverless cannot hold the models and the database.

So the live setup is **your PC plus a free Cloudflare quick tunnel**. A cloud VM (for example Oracle Cloud Always Free, which asks for a card for identity only) would stay online without your PC.

15.2 Going live

1. `backend\.env` has the Groq key and `JWT_SECRET`.
2. Double-click `scripts\share_public.bat`. It builds the website, starts the server in a minimised window ("Lumen server"), then starts `cloudflared` and prints `https://<words>.trycloudflare.com`.
3. Send that link. The login page offers one-click demo organizations.

How the link reaches your PC:

```
Visitor → https://....trycloudflare.com → Cloudflare → your cloudflared window → localhost:8000 → Lumen
```

Rules: keep the `cloudflared` window open and the PC on and online. Closing the tunnel ends the link; a new run gives a new address (and new assistant links).

15.3 Updating after a code change

Double-click `scripts\update_live.bat`. It runs `npm run build`, stops the old server on port 8000, starts the new one, and waits until it is healthy. The tunnel keeps running, so the public link stays the same. Then press **Ctrl+F5** in the browser.

By hand:

```
cd frontend && npm run build                       :: only if frontend files changed
:: stop the old server (close the "Lumen server" window, or free port 8000)
cd ..\backend && set SEED_DEMO=true
uv run uvicorn app.main:app --host 127.0.0.1 --port 8000 --proxy-headers --forwarded-allow-ips=*
```

Save to GitHub:

```
cd D:\ishita\app
git add -A
git commit -m "describe the change"
git push
```

The repository uses an SSH key (`~/.ssh/id_ed25519_lumen`, configured as `core.sshCommand` in that repo) added to the `salman-kv` GitHub account.

15.4 Before going public for real

- Delete the Groq key shared in chat and make a new one.
- Use a fresh long `JWT_SECRET`.
- Turn off `SEED_DEMO`.
- Use a real database and persistent storage (not a PC), HTTPS, and a persistent rate limiter (the current one is in memory).

16. Troubleshooting

Symptom	Cause and fix
Answers are short generic text, `degraded` is true, step says "template fallback (LLM unavailable)"	The language-model call is failing. Test: `https://api.groq.com/openai/v1/models` with your key. Usually a wrong `OPENAI_MODEL` name, a bad key, or no internet. Fix `.env` and restart
"You're asking too quickly"	Rate limit (30 per minute per token). Wait a minute. Set `RATE_LIMIT_PER_MINUTE` higher for load tests
`/api/health` shows `"ok": false` for the LLM	Same as the first row
Some answers are cut off mid-sentence or very short	A model was rate-limited. Groq's free tier allows about 8,000 tokens a minute per model; failover to `OPENAI_FALLBACK_MODELS` usually hides this. Under heavy testing, wait a minute, or use a paid key
Browser still shows the old site after an update	Hard refresh with **Ctrl+F5**. The build produces new file names, so a refresh picks them up
Port 8000 already in use	Another server is running. Close its window, or free the port (see `update_live.bat`)
First start is slow	Models (about 150 MB) download once into the cache
Documents stuck on "processing"	Check the server window for an error; the document normally turns `failed` with a reason. Replace the file or re-upload
"Upload failed: not a valid PDF / too many pages / too large"	Working as intended; limits are 10 PDFs, 20 pages, 10 MB. Scanned-image PDFs without text cannot be read (no OCR)

Symptom	Cause and fix
The public link stopped working	The tunnel window was closed or the PC slept. Start `share_public.bat` for a new link
Assistant link says "invalid or replaced"	The token was regenerated. Copy the new link from the Assistant tab
Eval or battery shows many failures at once	Rate limit, or wrong demo tokens (`.tokens.json` is stale). Run `seed_and_eval.py --seed-only`, wait a minute, run again
Login fails after resetting data	Deleted `backend/data`. Restart with `SEED_DEMO=true` to recreate the demo organizations, or register again
`git push` is rejected with 403	Wrong GitHub account saved. Use the SSH remote with the `lumen` key

Where to look: the "Lumen server" window (or `server.log` if started in the background) prints every request; `GET /api/health` summarises the state.

17. How to extend it

Add a new language-model provider. In `llm.py`, add a branch in `chat_json`, `stream_text` and `health`, and a value for `LLM_PROVIDER` in `config.py`. Anything speaking the OpenAI API needs no code: set the three `OPENAI_*` variables.

Add a new question type. Add example questions under a new key in `classifier.SEED`. It retrains at startup. Then use the label in `engine.py` where needed.

Teach it a new rule wording. Add a pattern in `rules.mine_page` and a test in `test_features.py` (`test_boolean_evaluator` shows the shape). Use `rules.describe` to control how the rule reads in answers.

Add a field the user can state (for example "department"). Add extraction in `engine.pattern_facts`, add its grounding in `ground_facts`, and make sure a rule card uses the same `field` name.

Add a dashboard page. Create `pages/dash/YourPage.jsx`, add it to `NAV` and the `Routes` in `Dashboard.jsx`, call your API through `api()`. Add a `data-tour` marker and a step in `Tour.jsx` if it should be in the walkthrough.

Add an endpoint. Add it in `main.py`. For admin endpoints depend on `current_org` and **filter every query by `org.id`**. Add a test that organization B cannot see organization A's data.

Change the database. Set `DATABASE_URL`. For new columns, update `db.py` (and the small migration in `init_db`) or introduce Alembic.

Rules for contributors: every query is tenant-scoped; the LLM never decides a verdict; every new behavior gets a test; run `uv run pytest -q` before pushing.

18. Known limits and next steps

Known limits (also stated in the README):

- Rule mining is pattern-based; unusual wordings can be missed. An LLM rule pass exists but is off by default.
- Decisions combine with **AND** only (no OR or weights).
- Boolean evaluators are deterministic comparators, not trained networks, by design.
- The assistant looks back only four turns.
- English only. No OCR for scanned PDFs.
- Rate limiter and sessions are in memory or stateless; fine for one process, not for many.
- The free setup depends on your PC being on.

Natural next steps:

- Host on an always-on VM with HTTPS and a real database (Postgres).
- Add an admin "gaps" report grouping unanswered questions by topic.
- Add feedback (thumbs up/down) to the history.
- Add OR conditions and priorities to rules, and a rule editor in the dashboard.
- OCR for scanned PDFs.
- A background worker queue for ingestion at scale.

19. Glossary

Term	Meaning
Tenant / organization	One customer. Their data is isolated from every other
Knowledge base (KB)	The organization's uploaded PDFs and everything derived from them
Assistant token	The secret in the public link (`ast_...`) that identifies which organization's assistant to talk to
JWT	A signed login token for the admin dashboard
Chunk / passage	A piece of a page (about 650 characters) that is searched and cited
Embedding	A list of 384 numbers that represents a passage's meaning, so similar meanings are close together
BM25	A classic keyword-matching score
RRF	Reciprocal Rank Fusion: merges two ranked lists into one
Reranker / cross-encoder	A model that reads question and passage together to judge how well the passage answers it
System 1	Small, fast, specialised models and rules that make the decisions (as opposed to a large language model)
Rule card	A typed condition mined from a PDF: benefit, field, operator, value, evidence, page
Boolean decision	Does a stated fact satisfy a rule: pass, fail or unknown
Grounding	Making sure an answer is supported by the documents; refusing when it is not
Hallucination	A model stating something that is not in the documents
SSE	Server-Sent Events: a one-way stream from server to browser, used to show the answer live
Tunnel	A secure path that gives a service running on your PC a public web address
SPA	Single-page application: the React site, one HTML file that draws every page

20. Command cheat sheet

```
:: ---- run ----
cd backend && uv sync
uv run uvicorn app.main:app --reload --port 8000        :: development API
cd frontend && npm install && npm run dev                 :: development website (port 5173)
cd frontend && npm run build                              :: production website build

:: ---- live ----
scripts\share_public.bat                                  :: build + start + public link
scripts\update_live.bat                                   :: rebuild + restart, same link
curl http://localhost:8000/api/health                     :: is it healthy?

:: ---- test ----
cd backend && uv run pytest -q                            :: 119 tests
uv run python ../scripts/seed_and_eval.py                 :: seed demo orgs + 19-question evaluation
uv run python ../scripts/scenario_battery.py              :: 44 scenarios

:: ---- docker ----
```

```
docker compose up --build                                   :: app + Ollama
docker compose down                                         :: stop

:: ---- git ----
git status
git add -A && git commit -m "message" && git push
```

This document is also available as a PDF: `docs/Lumen_Documentation.pdf`. After editing this file, regenerate it with `cd backend && uv run python ../scripts/build_docs_pdf.py`.